AI Playbook and Toolkit for Public Health Departments

Agentic Workflow Design and Orchestration

ARC 320

Agentic AI refers to AI systems configured to pursue a goal through multiple steps, often by using tools, calling APIs, retrieving information, creating tasks, or taking actions. In public health, agentic workflows may eventually support intake triage, document routing, surveillance review, resource referral, emergency operations, procurement review, or administrative automation. These uses require stronger controls than simple drafting or summarization. This module teaches learners how to design agentic workflows with clear boundaries. An AI agent should not be understood as an independent decision-maker. It is a software component operating within a governed workflow. Public health agencies must define the goal, permitted tools, data access, action limits, human approval points, logs, failure handling, and rollback procedures. Learners will create an agentic workflow control plan that identifies allowed actions, prohibited actions, oversight requirements, and incident triggers.

Level	intermediate/practitioner
Primary track	Technical Architecture Track
Appears in tracks	analytics-modeling

Learning Objectives

- Define agentic AI and orchestration in public health terms.
- Distinguish advisory outputs from tool-using or action-taking AI workflows.
- Identify the components of an agentic workflow, including goals, tools, permissions, memory, planning, human approval, logs, and rollback.
- Assess risks from excessive agency, unauthorized actions, error propagation, prompt injection, and unclear accountability.
- Design human-in-the-loop and human-on-the-loop controls for public health workflows.
- Produce an agentic workflow control plan.

Module Overview

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Why This Topic Matters for Public Health AI

Agentic AI matters because it shifts AI from generating outputs to participating in workflows. A chatbot that drafts a message is one level of risk. An agent that sends the message, updates a case record, creates a task, or notifies a partner is a different level of risk. Action-taking systems require explicit governance because errors can propagate quickly.

Public health workflows often involve legal authority, sensitive data, vulnerable communities, and time-sensitive decisions. An agentic workflow could be valuable if it helps route records, gather relevant documents, prepare draft summaries, or create follow-up tasks. It could be harmful if it acts outside scope, updates records incorrectly, exposes information, or creates alerts without appropriate review.

Responsible agentic design begins with constraint. Agencies should define what the agent is allowed to know, allowed to do, required to ask, required to log, and prohibited from doing. The system should fail safely when it encounters uncertainty, missing data, conflicting guidance, or requests outside approved scope.

Public Health Example

A health department may consider an agentic workflow for incoming public health inquiries. The agent could classify the inquiry, retrieve relevant approved guidance, draft a response, create a ticket, and route the item to the appropriate program. For low-risk general information, the agent might prepare a draft for staff review. For sensitive complaints, reportable disease concerns, media inquiries, or legal issues, the agent should escalate rather than respond.

This design requires orchestration rules. The agent needs classification criteria, approved guidance sources, limits on data access, prohibited response categories, human approval before sending messages, audit logs, and procedures for correcting misrouted items. The value comes from workflow support, not from removing accountability.

Technical Deep Dive

Agentic workflow design begins with the goal. The goal should be narrow and operationally defined. “Help with surveillance” is too broad. “Draft a summary of new records and create a review task for a supervisor when defined criteria are met” is more suitable. Narrow goals make it easier to define tools, permissions, validation, and monitoring.

Tool access is a central design decision. An agent may be allowed to search documents, query a database, create a task, send a draft message, or update a record. Each tool should be reviewed for sensitivity, consequence, and reversibility. Read-only access is generally lower risk than write access. External communication and system-of-record updates usually require human approval.

Memory should be handled carefully. Some agents retain context across interactions or workflows. Persistent memory may improve continuity but can create privacy, accuracy, and records risks. Agencies should define what may be stored, for how long, who can access it, and how it can be corrected or deleted.

Human oversight should match risk. Human-in-the-loop approval is appropriate for actions that affect records, services, public communication, legal obligations, or partners. Human-on-the-loop monitoring may be appropriate for lower-risk background tasks if users can intervene quickly. Fully automated action should be limited to low-risk, well-tested, reversible tasks.

Agentic workflows require incident response planning. Agencies should define what counts as an agentic incident, such as unauthorized action, data exposure, repeated error, misrouting, unsafe recommendation, or failure to escalate. Logs should support reconstruction of prompts, retrieved sources, tool calls, actions, approvals, and outcomes.

Risks, Failure Modes, and Guardrails

Excessive agency. An agent may take actions beyond intended scope. Guardrails include narrow goals, allowed-action lists, prohibited actions, and approval gates.

Unauthorized system access. Tool access may expose sensitive data or functions. Guardrails include least privilege, scoped credentials, role-based access, and audit logs.

Error propagation. One wrong step may trigger downstream errors. Guardrails include checkpoints, validation, rollback, and staged deployment.

Unclear accountability. Staff may not know who is responsible for agent actions. Guardrails include role definitions, review records, and governance ownership.

Application to AI-Supported Workflows

Agentic workflows may combine RAG, APIs, prompts, classification models, workflow rules, and messaging systems. They are most appropriate when tasks are repetitive, bounded, reviewable, and supported by reliable data. They are least appropriate when the task requires complex judgment, legal interpretation, community sensitivity, or irreversible action without review.

Public health agencies should pilot agentic workflows in limited, low-risk settings before expanding. Monitoring should include tool calls, failed actions, overrides, user feedback, time saved, error types, and unintended consequences.

Reflection Questions

What goal would the agent pursue, and how narrowly can it be defined?

Which tools or systems would the agent access?

Which actions are allowed, prohibited, or require human approval?

What logs are needed to reconstruct agent behavior?

What rollback or correction process is needed if the agent makes an error?

Practical Exercise

Create an agentic workflow control plan. Include the goal, trigger, data sources, tools, allowed actions, prohibited actions, approval points, memory rules, logs, monitoring indicators, rollback process, incident triggers, and responsible owner.

The plan should identify at least one low-risk pilot boundary and one condition that would require pausing the workflow.

Expected Artifact or Evidence

Agentic workflow control plan

Allowed/prohibited action list

Human approval and escalation map

Tool-call logging requirements

Expected Assignment

- Agentic workflow control plan
- Allowed/prohibited action list
- Human approval and escalation map
- Tool-call logging requirements

Knowledge Check

- 1. What makes agentic AI different from simple drafting support?
- 2. Which action usually requires stronger oversight?
- 3. What is orchestration?
- 4. What should happen when an agent encounters an out-of-scope request?
- 5. What is strong completion evidence for this module?

References and Resources

- OWASP Top 10 for Large Language Model Applications: https://owasp.org/www-project-top-10-for-large-language-model-applications/
- NIST AI RMF Generative AI Profile: https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence

- NIST Artificial Intelligence Risk Management Framework: https://www.nist.gov/itl/ai-risk-management-framework
- NIST Privacy Framework: https://www.nist.gov/privacy-framework